

Saber Ayatollahi, CFA, MSc

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SUMMARY

Associate Director-level ALM and balance-sheet risk specialist with ~7 years across IRRBB analytics, multi-scenario cash flow projection, and hedging analysis for institutional portfolios. At Moody's Analytics, holds delegated sign-off authority on valuation, sensitivity, and ALM outputs for portfolios of \$5-25bn per engagement, including review under parallel and non-parallel rate shocks aligned with banking book IRRBB standards. Prior pension ALM, LDI, and interest-rate / inflation / currency hedging delivery at Ortec Finance. CFA charterholder; dual MSc; advanced Python and SQL.

CORE SKILLS

ALM & IRRBB – EVE / NII, parallel & non-parallel shocks • Structural interest rate risk & duration gap analytics • Balance sheet risk modelling & banking book analytics • Fixed income & interest rate derivatives valuation • Yield curve construction, calibration & key-rate duration • Multi-scenario cash flow projection & liquidity gap modelling • Hedging strategy analysis (interest rate, inflation, FX) • Stochastic scenario generation & Monte Carlo simulation • Python (pandas, NumPy, SciPy), SQL, Excel • Senior-stakeholder & ALCO-ready analytical reporting

EXPERIENCE

Moody's Analytics

Asst. Director – Modelling Services (Investment & ALM Risk Lead) | Toronto, May 2023 - Present

Structural interest rate risk & ALM analytics

- Hold delegated sign-off authority on valuation, sensitivity, and ALM outputs for multi-asset institutional portfolios of \$5-25bn per engagement (~\$50bn+ cumulative), independently reviewing curve construction, spread calibration, and cross-asset interactions prior to production release.
- Oversee interest-rate risk and duration analysis under parallel and non-parallel rate shocks across the banking book, aligned with IRRBB standards analogous to OSFI B-12 and Basel Committee frameworks.
- Review derivatives pricing outputs across rates, FX, and inflation, verifying consistency of sensitivities and scenario impacts at portfolio-level ALM aggregates that inform hedging decisions.

Cash flow projection & balance-sheet modelling

- Led design and implementation of an enterprise multi-asset cash flow projection engine supporting base, stress, and reverse-stress scenarios, delivering forward-looking liquidity and balance sheet visibility across ALM horizons.
- Architected configurable time-bucketed liquidity gap analytics (T+1 through multi-year) enabling funding optimization, refinancing risk assessment, and asset-allocation decisions.
- Embedded behavioral cash flow assumptions, prepayment logic, and macro stress overlays aligned with regulatory liquidity expectations.

Governance, tooling & stakeholder reporting

- Operate within a formal model-governance framework; escalate outputs lacking mathematical or economic defensibility; serve as primary escalation point for client-critical analytical issues, interfacing directly with Heads of Risk and investment stakeholders.
- Re-engineered manual spreadsheet workflows into scalable Python analytics pipelines with embedded logging, validation, and auditability controls.
- Prepare analytical summaries for senior stakeholders covering interest-rate exposure, scenario impacts, and balance-sheet sensitivities, translating quantitative outputs into committee-ready narratives.

Client Service Specialist – PFaroe Platform | Toronto, May 2022 - May 2023

- Led onboarding and implementation for U.S. and Canadian pension funds, asset managers, and consulting firms migrating to the PFaroe analytics platform; advised on asset, liability, and ALM configurations and validated model outputs post-deployment.

- Served as technical liaison between client investment teams and Product Owners, scoping requirements into structured development requests; contributed to successful migration of all assigned client accounts from legacy Calypso to PFaroe PM.

Ernst & Young (EY)

Senior Consultant – IFRS & Financial Risk Transformation | Toronto, Sep 2021 – Apr 2022

- Delivered IFRS 17 and IFRS 9 transformation projects for Canadian and international insurance clients, aligning regulatory requirements with system implementation and financial-reporting frameworks.
- Advised actuarial, IT, finance, and program-management functions on adoption of IFRS 17 and IFRS 9 standards; contributed to governance documentation and regulatory-readiness processes.

Ortec Finance Canada

Consultant – Asset-Liability Management & Risk | Toronto, Feb 2019 – Sep 2021

- Delivered ALM studies for Canadian and international pension funds and institutional investors, analyzing funding ratios, duration mismatch, and long-term balance-sheet risk exposures.
- Built and interpreted stochastic economic scenario generators assessing interest-rate risk and funding volatility under base-case and stressed environments.
- Conducted liability-driven investment (LDI) analysis, evaluating trade-offs between return objectives, duration alignment, and liquidity considerations for pension-fund clients.
- Performed cash flow, liquidity, and hedging analysis across interest-rate, inflation, and currency risk for international institutional clients.
- Advised clients on Strategic and Tactical Asset Allocation trade-offs via stress testing and sensitivity analysis; presented findings to investment committees on-site.
- Led onboarding and ALM study for a three-plan university pension merger (UPP / JSPP), developing and validating the model capturing Funding Policy dynamics and evaluating duration, currency, inflation, and leverage overlay strategies.

EDUCATION & CREDENTIALS

- Chartered Financial Analyst (CFA) – CFA Institute, 2024
- MSc, Financial Modelling – Western University, 2018
- MSc, Chemical Engineering (Full Scholarship) – Western University, 2016